1.31 Differentiation: Introduction

Worked Solutions

1a)

$$\frac{dy}{dx} = 12x^2 + 10x$$

1b)

$$\frac{dy}{dx} = 2 - 18x$$

1c)

$$\frac{dy}{dx} = 6x^5 - 36x$$

1d)

$$\frac{dy}{dx} = -x^{-2} + 4x^3$$

1e)

$$\frac{dy}{dx} = 3x^2 + 2x^{-3}$$

1f)

$$\frac{dy}{dx} = 2x - 2x^{-3} + 1$$

1g)

$$\frac{dy}{dx} = 15x^4 + 4x^{-3} - 1$$

1h)

$$\frac{dy}{dx} = 2ax^{a-1} - bx^{b-1}$$

2a)

$$f'(t) = t - 3t^{-6}$$

2b)

$$f'(t) = \frac{1}{4} - 3t^3$$

2c)

$$f'(t) = \frac{1}{2}t^5 + 20t$$

2d)

$$f'(t) = -\frac{1}{3}t^{-2} + 4t^5$$

2e)

$$f'(t) = \frac{7}{4}t + \frac{3}{4}t^{-4}$$

2f)

$$f'(t) = \frac{16}{7}t^7 - \frac{3}{7}t^{-4} + \frac{1}{5}$$

2g)

$$f'(t) = -\frac{3}{2}t^5 + t^{-3} + 2$$

2h)

$$f'(t) = -\frac{a}{2b}t^{a/b-1}$$

3a)

$$\frac{dy}{dt} = 4t^{-1/2} + 2t^{-2/3}$$

3b)

$$\frac{dy}{dt} = 5 - 3t^{-3/4}$$

3c)

$$\frac{dy}{dt} = \frac{2}{3}t^{-1/3} + \frac{21}{5}t^{-2/5}$$

3d)

$$\frac{dy}{dt} = -\frac{1}{2}t^{-3/2} + \frac{1}{4}t^{-3/4}$$

3e)

$$\frac{dy}{dt} = -\frac{5}{8}t^{-3/8} + \frac{3}{8}t^{-11/8} - \frac{1}{8}$$

3f)

$$\frac{dy}{dt} = \frac{1}{3}t^{-2/3} - \frac{1}{3}t^{-4/3} + 3$$

3g)

$$\frac{dy}{dt} = -\frac{3}{4}t^{-1/4} - \frac{8}{9}t^{-17/9} - \frac{1}{12}$$

4a)

$$f(x) = x^2 + 3x \text{ so } f'(x) = 2x + 3$$

4b)

$$f(x) = x^2 + 6x + 5 \text{ so } f'(x) = 2x + 6$$

4c)

$$f(x) = 2x^2 + x - 10 \text{ so } f'(x) = 4x + 1$$

4d)

$$f(x) = x^2 + 8x + 16 \text{ so } f'(x) = 2x + 8$$

4e)

$$f(x) = x^3 - x^2 \text{ so } f'(x) = 3x^2 - 2x$$

4f)

$$\begin{aligned} f(x) &= (x^2 + 4x + 4)(5x - 1) \\ &= 5x^3 + 19x^2 + 16x + 16 \\ \text{so } f'(x) &= 15x^2 + 38x + 16 \end{aligned}$$

4g)

$$f(x) = 7x^2 - x^4 \text{ so } f'(x) = 14x - 4x^3$$

4h)

$$f(x) = 9x^3 - x \text{ so } f'(x) = 27x^2 - 1$$

4i)

$$f(x) = 7 - x^{-1} + 8x \text{ so } f'(x) = x^{-2} + 8$$

4j)

$$\begin{aligned} f(x) &= \left(x + \frac{1}{x}\right)(x^2 + 2 + x^{-2}) \\ &= x^3 + 3x + 3x^{-1} + x^{-3} \\ \text{so } f'(x) &= 3x^2 + 3 - 3x^{-2} - 3x^{-4} \end{aligned}$$

4k)

$$\begin{aligned} f(x) = (x+p)(x+q) &= x^2 + (p+q)x + pq \\ \text{so } f'(x) &= 2x + p + q \end{aligned}$$

4l)

$$\begin{aligned} f(x) = (x-a)\left(x + \frac{1}{a}\right) &= x^2 - ax + \frac{1}{a}x - 1 \\ &= x^2 + \frac{1-a^2}{a}x - 1 \\ \text{so } f'(x) &= 2x + \frac{1-a^2}{a} \end{aligned}$$

5a)

$$x^{1/2}$$

5b)

$$x^{5/2}$$

5c)

$$x^{-4}$$

5d)

$$x^3 x^{3/2} = x^{9/2}$$

5e)

$$x^{1/3}$$

5f)

$$x^{-1/2}$$

5g)

$$x^1 x^{-1/2} = x^{1/2}$$

5h)

$$x^{-1} x^{-1/2} = x^{-3/2}$$

5i)

$$x^{-1} x^{-1/a} = x^{-1-1/a} = x^{-(a+1)/a}$$

6a)

$$5x^{1/3}$$

6b)

$$2x^{3/2}$$

6c)

$$3x^{-2}$$

6d)

$$2^5 x^5 x^{5/2} = 32x^{15/2}$$

6e)

$$\frac{1}{3}x^{1/8}$$

6f)

$$4x^{-1/2}$$

6g)

$$3x^1 3^{-1} x^{-4/2} = x^{-1}$$

6h)

$$5x^{-1}4^{-1}x^{-1/2} = \frac{5}{4}x^{-3/2}$$

6i)

$$2 \times 5^{-1} \times a^{-1/2} \times x^{-1/2} = \frac{2}{5\sqrt{a}}x^{-1/2}$$

7a)

$$y = x^{1/4} \text{ so } \frac{dy}{dx} = \frac{1}{4}x^{-3/4}$$

7b)

$$y = 3x^{5/2} \text{ so } \frac{dy}{dx} = \frac{15}{2}x^{3/2}$$

7c)

$$y = 5x^{-3} \text{ so } \frac{dy}{dx} = -15x^{-4}$$

7d)

$$y = x^3 x^{3/2} = x^{9/2} \text{ so } \frac{dy}{dx} = \frac{9}{2}x^{7/2}$$

7e)

$$y = \frac{1}{5}x^{1/3} \text{ so } \frac{dy}{dx} = \frac{1}{15}x^{-2/3}$$

7f)

$$y = 3x^{-1/2} \text{ so } \frac{dy}{dx} = -\frac{3}{2}x^{-3/2}$$

7g)

$$y = x^1 x^{-2} = x^{-1} \text{ so } \frac{dy}{dx} = -x^{-2}$$

7h)

$$y = 2x4^{-1}x^{-2} = \frac{1}{2}x^{-1} \text{ so } \frac{dy}{dx} = -\frac{1}{2}x^{-2}$$

8a)

$$f(x) = x^{3/2} + 2x^{1/2} \text{ so } f'(x) = \frac{3}{2}x^{1/2} + x^{-1/2}$$

8b)

$$f(x) = x^{5/2} - x^{1/2} \text{ so } f'(x) = \frac{5}{2}x^{3/2} - \frac{1}{2}x^{-1/2}$$

8c)

$$f(x) = x^2 + 4 \text{ so } f'(x) = 2x$$

8d)

$$f(x) = 2x^2 + 4x^{-3} \text{ so } f'(x) = 4x - 12x^{-4}$$

8e)

$$f(x) = 9x^{-1} - x^{-2} \text{ so } f'(x) = -9x^{-2} + 2x^{-3}$$

8f)

$$f(x) = x^{3/2} + 2x^{1/2} - x^{-1/2}$$
$$\text{so } f'(x) = \frac{3}{2}x^{1/2} + x^{-1/2} + \frac{1}{2}x^{-3/2}$$

8g)

$$f(x) = x^{3/2} - 5x^{-2} + x^{-1/2}$$
$$\text{so } f'(x) = \frac{3}{2}x^{1/2} + 10x^{-3} - \frac{1}{2}x^{-3/2}$$

9a)

$$\frac{dy}{dx} = 3x^2 - 3x^{-4} + 4x - 5, \quad \frac{d^2y}{dx^2} = 6x + 12x^{-5} + 4$$

9b)

$$\frac{dy}{dx} = \frac{1}{3}x^3 + \frac{1}{4}x + 7, \quad \frac{d^2y}{dx^2} = x^2 + \frac{1}{4}$$

9c)

$$\frac{dy}{dx} = -3x^2 + x^{-3/2} + 15, \quad \frac{d^2y}{dx^2} = -6x - \frac{3}{2}x^{-5/2}$$

9d)

$$\frac{dy}{dx} = -\frac{1}{7}x^{-6/7} + \frac{1}{7}x^{-8/7} - \frac{1}{9}, \quad \frac{d^2y}{dx^2} = \frac{6}{49}x^{-13/7} - \frac{8}{49}x^{-15/7}$$

9e)

$$y = 2x^4 + x \text{ so } \frac{dy}{dx} = 8x^3 + 1, \quad \frac{d^2y}{dx^2} = 24x^2$$

9f)

$$y = 4x^{3/2} \text{ so } \frac{dy}{dx} = 6x^{1/2}, \quad \frac{d^2y}{dx^2} = 3x^{-1/2}$$

9g)

$$y = 2x^{-1/2} \text{ so } \frac{dy}{dx} = -x^{-3/2}, \quad \frac{d^2y}{dx^2} = \frac{3}{2}x^{-5/2}$$

9h)

$$y = x^4 + 4x^{-3} \text{ so } \frac{dy}{dx} = 4x^3 - 12x^{-4}, \quad \frac{d^2y}{dx^2} = 12x^2 + 48x^{-5}$$

10a)

$$\frac{dy}{dx} = 2x + 5 \text{ so } \left.\frac{dy}{dx}\right|_{x=1} = 7$$

10b)

$$\frac{dy}{dx} = 3x^2 - 2 \text{ so } \left.\frac{dy}{dx}\right|_{x=-2} = 10$$

10c)

$$y = x^{-1/2} - 2x^2$$
$$\frac{dy}{dx} = -\frac{1}{2}x^{-3/2} - 4x \text{ so } \left.\frac{dy}{dx}\right|_{x=4} = -\frac{1}{16} - 16 = -\frac{257}{16}$$

10d)

$$\frac{dy}{dx} = \frac{1}{2}x^{-1/2} - 1 \text{ so } \left.\frac{dy}{dx}\right|_{x=4} = \frac{1}{4} - 1 = -\frac{3}{4}$$

10e)

$$y = x^3 - 3x^2 \text{ so } \frac{dy}{dx} = 3x^2 - 6x$$
$$\left.\frac{dy}{dx}\right|_{x=2} = 12 - 12 = 0$$

11a)

$$f'(x) = 3x^2 - 10x \text{ so } f''(x) = 6x - 10$$
$$f''(1) = -4$$

11b)

$$f(x) = x^{-2} - x^2$$
$$f'(x) = -2x^{-3} - 2x \text{ so } f''(x) = 6x^{-4} - 2$$
$$f''(1) = 4$$

11c)

$$f(x) = x^{1/2} + 2x^2$$
$$f'(x) = \frac{1}{2}x^{-1/2} + 4x \text{ so } f''(x) = -\frac{1}{4}x^{-3/2} + 4$$
$$f''(1) = \frac{15}{4}$$

11d)

$$f(x) = x^{4/3} - 4x$$
$$f'(x) = \frac{4}{3}x^{1/3} - 4 \text{ so } f''(x) = \frac{4}{9}x^{-4/3}$$
$$f''(1) = \frac{4}{9}$$

11e)

$$f(x) = px^{1/2} - \frac{4}{q}x^{-1}$$
$$f'(x) = \frac{1}{2}px^{-1/2} + \frac{4}{q}x^{-2} \text{ so } f''(x) = -\frac{1}{4}px^{-3/2} - \frac{8}{q}x^{-3}$$
$$f''(1) = -\frac{1}{4}p - \frac{8}{q}$$

12a)

$$f'(x) = 2e^{2x}$$

12b)

$$f'(x) = 13e^{13x}$$

12c)

$$f'(x) = -8e^{-8x}$$

12d)

$$f'(x) = \frac{1}{5}e^{x/5}$$

12e)

$$f'(x) = -\frac{4}{7}e^{-4x/7}$$

12f)

$$f'(x) = 6e^{3x}$$

12g)

$$f'(x) = -5e^{-x}$$

12h)

$$f'(x) = -\frac{1}{2}e^{-2x}$$

12i)

$$f(x) = e^{-5x} \text{ so } f'(x) = -5e^{-5x}$$

12j)

$$f(x) = e^{3x/2} \text{ so } f'(x) = \frac{3}{2}e^{3x/2}$$

12k)

$$f(x) = e^{x}e^{-x/2} = e^{x/2} \text{ so } f'(x) = \frac{1}{2}e^{x/2}$$

12l)

$$f(x) = e^{x}e^{-3x} = e^{-2x} \text{ so } f'(x) = -2e^{-2x}$$

12m)

$$f(x) = e^{ax/2} \text{ so } f'(x) = \frac{a}{2}e^{ax/2}$$

12n)

$$f(x) = e^{x}e^{-ax/2} = e^{(1-a/2)x} \text{ so } f'(x) = \left(1 - \frac{a}{2}\right)e^{(1-a/2)x}$$

13a)

$$\frac{dy}{dx} = 4e^{2x} - 15e^{3x}$$

13b)

$$\frac{dy}{dx} = 2 - 3e^{3x}$$

13c)

$$\frac{dy}{dx} = -e^{-x} + 5x^4$$

13d)

$$\frac{dy}{dx} = 5x^4 - x^3 - \frac{3}{5}e^{-x/5}$$

13e)

$$y = x^{1/2} + 3e^{-2x} \text{ so } \frac{dy}{dx} = \frac{1}{2}x^{-1/2} - 6e^{-2x}$$

13f)

$$y = \sqrt{a}x^{1/2} + 5e^{-bx} \text{ so } \frac{dy}{dx} = \frac{\sqrt{a}}{2}x^{-1/2} - 5be^{-bx}$$

14a)

$$f'(x) = 3e^{3x} - 8e^{-x} \text{ so } f''(x) = 9e^{3x} + 8e^{-x}$$

14b)

$$f'(x) = 10x - 5e^{5x} \text{ so } f''(x) = 10 - 25e^{5x}$$

14c)

$$f'(x) = -2e^{-x} + 2x \text{ so } f''(x) = 2e^{-x} + 2$$

14d)

$$f'(x) = 4x^3 + 4e^{-x/2} + \frac{3}{2}x^{-4}$$
$$\text{so } f''(x) = 12x^2 - 2e^{-x/2} - 6x^{-5}$$

15a)

$$\frac{dy}{dx} = e^x \text{ so } \left.\frac{dy}{dx}\right|_{x=2} = e^2$$

15b)

$$\frac{dy}{dx} = 5e^x \text{ so } \left.\frac{dy}{dx}\right|_{x=-1} = 5e^{-1}$$

15c)

$$\frac{dy}{dx} = 2x - e^x \text{ so } \left.\frac{dy}{dx}\right|_{x=0} = 0 - 1 = -1$$

15d)

$$\frac{dy}{dx} = 1 - e^x, \quad \left.\frac{dy}{dx}\right|_{x=1} = 1 - e$$

15e)

$$y = 2x + 2e^{2x} \text{ so } \frac{dy}{dx} = 2 + 4e^{2x}$$
$$\left.\frac{dy}{dx}\right|_{x=1/2} = 2 + 4e$$

15f)

$$y = -3x - 3e^{ax} \text{ so } \frac{dy}{dx} = -3 - 3ae^{ax}$$
$$\left.\frac{dy}{dx}\right|_{x=1} = -3 - 3ae^{a}$$

16)

$$y = 3x - 3e^x + 5 \text{ so } \frac{dy}{dx} = 3 - 3e^x.$$
$$\left.\frac{dy}{dx}\right|_{x=-1.423} = 3 - 3e^{-1.423} = 2.28 \text{ (2 d.p.)}$$
$$\left.\frac{dy}{dx}\right|_{x=0} = 3 - 3e^{0} = 0$$
$$\left.\frac{dy}{dx}\right|_{x=0.969} = 3 - 3e^{0.969} = -4.91 \text{ (2 d.p.)}$$

17)

$$y = ke^{-px}, \quad \frac{dy}{dx} = -pke^{-px}, \quad \frac{d^2y}{dx^2} = p^2ke^{-px}.$$

$e^{-px} > 0$ for all values of p and x.

If $p = 0$ or $k = 0$ then $p^2ke^{-px} = 0$.

If $p \neq 0$ and $k > 0$, then $p^2ke^{-px} > 0$.

If $p \neq 0$ and $k < 0$, then $p^2ke^{-px} < 0$.

So $\dfrac{d^2y}{dx^2} > 0$ when $k > 0$ and $p \neq 0$.